Physics 4B Spring 2026
Week 10 Discussion

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1. **Brainwarmer**[1]. Find the change in entropy if 500 g of water at 80°C is added to 300 g of water at 20°C.

   Solution: Mixing the two parts of water will result in a temperature in between the two initial temperatures. We can add the entropy changes for the two parts together to get the total entropy change.

   First, we have to determine the final temperature

   $$\begin{aligned} 0 = \Delta Q &= m_1 c_\mathrm{w} \Delta T_1 + m_2 c_\mathrm{w} \Delta T_2 \\ &= m_1 c_\mathrm{w} (T_f - T_{1i}) + m_2 c_\mathrm{w} (T_f - T_{2i}) \end{aligned} \tag{1}$$

   which leads to

   $$T_f = \frac{m_1 T_{1i} + m_2 T_{2i}}{m_1 + m_2} = \frac{(0.5\text{ kg})(353\text{ K}) + (0.3\text{ kg})(293\text{ K})}{0.8\text{ kg}} = 330.5\text{ K} \tag{2}$$

   For the 500 g of water, the entropy change is

   $$\begin{aligned} \Delta S_1 &= \int \frac{dQ}{T} = m_1 c_\mathrm{w} \int_{T_{1i}}^{T_f} \frac{dT}{T} = m_1 c_\mathrm{w} \ln\left(\frac{T_f}{T_{1i}}\right) \\ &= (0.5\text{ kg})(4184\text{ J/(kg}\cdot\text{K)}) \ln\left(\frac{330.5\text{ K}}{353\text{ K}}\right) = -137.8\text{ J/K} \end{aligned} \tag{3}$$

   For the 300 g of water, the entropy change is

   $$\Delta S_2 = m_2 c_\mathrm{w} \ln\left(\frac{T_f}{T_{2i}}\right) = (0.3\text{ kg})(4184\text{ J/(kg}\cdot\text{K)}) \ln\left(\frac{330.5\text{ K}}{293\text{ K}}\right) = 151.2\text{ J/K} \tag{4}$$

   The total entropy change is therefore

   $$\Delta S = \Delta S_1 + \Delta S_2 = 13.4\text{ J/K} \tag{5}$$

2. **Carnot engines in series**[2]. Consider two Carnot engines in series: the first absorbs heat at T_1 and rejects some at T_0, and the second absorbs all that rejected heat at T_0 and rejects heat at T_2.

[1] Shankar 11.30
[2] Shankar 11.32

(i) What is η_{12}, the efficiency of the combination in terms of T_1, T_0, and T_2?

(ii) Express η_{12} in terms of η_{10} and η_{02}.

Solution:

(i) The efficiency of the combination does not care about the intermediate step, and would simply depend on the lowest and highest temperatures

$$\eta_{12} = 1 - \frac{T_2}{T_1} \tag{6}$$

(ii) We can think of the second engine as taking all the wasted heat from the first engine as the input heat, and converts part of that into work, so

$$\eta_{12} = \eta_{10} + (1 - \eta_{10})\eta_{02} \tag{7}$$

3. **Entropy of mixture**[3]. I mix 10 kg of Cu ($C_{\text{Cu}} = 386$ J/(kg $\cdot$ K) at 100°C) with 5 L of water ($C_{\text{w}} = 4180$ J/(kg $\cdot$ K)) at 0°C). What is

(i) the final T,

(ii) S_{Cu},

(iii) S_{w}, and

(iv) S_{universe}?

Solution:

(i) This is similar to Problem 1 except that the specific heats do not cancel out. 5 L of water has a mass of 5 kg. The final temperature is

$$\begin{aligned} T_f &= \frac{m_{\text{Cu}}c_{\text{Cu}}T_{\text{Cu},i} + m_{\text{w}}c_{\text{w}}T_{\text{w},i}}{m_{\text{Cu}}c_{\text{Cu}} + m_{\text{w}}c_{\text{w}}} \\ &= \frac{(10\text{ kg})(386\text{ J/(kg}\cdot\text{K)})(373\text{ K}) + (5\text{ kg})(4180\text{ J/(kg}\cdot\text{K)})(273\text{ K})}{(10\text{ kg})(386\text{ J/(kg}\cdot\text{K)}) + (5\text{ kg})(4180\text{ J/(kg}\cdot\text{K)})} \\ &= 288.6\text{ K} \end{aligned} \tag{8}$$

(ii) This is also similar to Problem 1. The entropy change in copper is

$$\begin{aligned} \Delta S_{\text{Cu}} &= m_{\text{Cu}}c_{\text{Cu}} \ln\left(\frac{T_f}{T_{\text{Cu},i}}\right) \\ &= (10\text{ kg})(386\text{ J/(kg}\cdot\text{K)}) \ln\left(\frac{288.6\text{ K}}{373\text{ K}}\right) = -990\text{ J/K} \end{aligned} \tag{9}$$

[3] Shankar 11.35

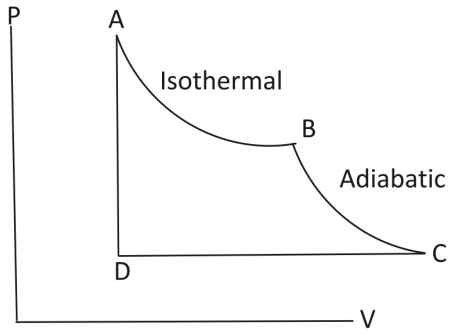
(iii) This is also similar. The entropy change in water is

$$\begin{aligned}\Delta S_{\mathrm{w}} &= m_{\mathrm{w}} c_{\mathrm{w}} \ln\left(\frac{T_f}{T_{\mathrm{w},i}}\right) \\ &= (5\ \mathrm{kg})(4180\ \mathrm{J/(kg\cdot K)}) \ln\left(\frac{288.6\ \mathrm{K}}{273\ \mathrm{K}}\right) = 1161\ \mathrm{J/K} \end{aligned} \tag{10}$$

(iv) The entropy change of the universe is the sum of the entropy changes of its subparts

$$\Delta S_{\mathrm{universe}} = \Delta S_{\mathrm{Cu}} + \Delta S_{\mathrm{w}} = 171\ \mathrm{J/K} \tag{11}$$

4. **Everything about heat engine**[4]. Consider 2 moles of an ideal diatomic gas taken through the cycle ABCDA shown in the figure below. You are given $T_A = 800$ K, $P_A = 6 \times 10^5$ Pa, $V_B = 3V_A$, $P_C = 10^5$ Pa, AB is isothermal, BC is adiabatic, CD is at constant pressure, and DA is at constant volume.



(i) Find (P, V, T) for all the corners.

(ii) Find the work done in each part of the cycle and the total.

(iii) Find the heat added in each part.

(iv) Find the efficiency η, and

(v) the entropy change in each part and the total.

Solution:

(i) For A, P_A and T_A are already given. The volume is

$$V_A = \frac{nRT_A}{P_A} = \frac{(2\ \mathrm{mol})(8.314\ \mathrm{J/(mol\cdot K)})(800\ \mathrm{K})}{6 \times 10^5\ \mathrm{Pa}} = 0.022\ \mathrm{m}^3 \tag{12}$$

[4] Adapted from Shankar 11.36

For B, since AB is an isotherm, $T_B = T_A = 800$ K. $V_B = 3V_A = 0.067$ m^3. The pressure is

$$P_B = \frac{nRT_B}{V_B} = \frac{1}{3}P_A = 2 \times 10^5 \text{ Pa} \tag{13}$$

For C, P_C is given. Since BC is adiabatic, with $\gamma = c_p/c_v = 7/5$, the volume is

$$V_C = \left(\frac{P_B V_B^\gamma}{P_C}\right)^{1/\gamma} = \left(\frac{(2 \times 10^5 \text{ Pa})(0.067 \text{ m}^3)^{7/5}}{10^5 \text{ Pa}}\right)^{5/7} = 0.11 \text{ m}^3 \tag{14}$$

and the temperature is

$$T_C = \frac{P_C V_C}{nR} = \frac{(10^5 \text{ Pa})(0.11 \text{ m}^3)}{(2 \text{ mol})(8.314 \text{ J/(mol} \cdot \text{K)})} = 656 \text{ K} \tag{15}$$

For D, $P_D = P_C = 10^5$ Pa and $V_D = V_A = 0.022$ m^3. The temperature is

$$T_D = \frac{P_D V_D}{nR} = \frac{(10^5 \text{ Pa})(0.022 \text{ m}^3)}{(2 \text{ mol})(8.314 \text{ J/(mol} \cdot \text{K)})} = 133 \text{ K} \tag{16}$$

(ii) For AB, the work done is

$$W_{AB} = \int_A^B P \, dV = nRT_A \int_{V_A}^{V_B} \frac{dV}{V} = nRT_A \ln\left(\frac{V_B}{V_A}\right) = 14614 \text{ J} \tag{17}$$

For BC, $\delta Q = 0$ implies $\delta W = -dU$, so the work done is

$$W_{BC} = -\Delta U_{BC} = \frac{5}{2}nR(T_B - T_C) = 5975 \text{ J} \tag{18}$$

This is the easiest way to find the work done in an adiabatic process, but I'll show the other way here as well. Using $PV^\gamma = $ const. along the BC path, we can write

$$\begin{aligned} W_{BC} &= \int_B^C P \, dV = P_B V_B^\gamma \int_{V_B}^{V_C} \frac{dV}{V^\gamma} = \frac{P_B V_B^\gamma}{1-\gamma}\left(V_C^{1-\gamma} - V_B^{1-\gamma}\right) \\ &= \frac{P_B V_B - P_B V_B^\gamma V_C^{1-\gamma}}{\gamma - 1} = \frac{P_B V_B - P_C V_C}{\gamma - 1} = 5975 \text{ J} \end{aligned} \tag{19}$$

For CD, the pressure is constant, so the work done is

$$W_{CD} = P_C \Delta V_{CD} = P_C(V_D - V_C) = -8695 \text{ J} \tag{20}$$

For DA, $\Delta V = 0$, so no work is done.

The total work done by the gas is

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA} = 11894 \text{ J} \tag{21}$$

(iii) For AB, the process is isothermal ($\Delta U = 0$), so

$$Q_{AB} = W_{AB} = 14614 \text{ J} \tag{22}$$

For BC, the process is adiabatic, so $Q_{BC} = 0$.

For CD, the heat into the system is

$$Q_{CD} = \Delta U_{CD} + W_{CD} = \frac{5}{2} nR(T_D - T_C) + W_{CD} = -30434 \text{ J} \tag{23}$$

For DA, the process is isochoric, so $Q_{DA} = \Delta U_{DA}$. The heat into the system is

$$Q_{DA} = \Delta U_{DA} = \frac{5}{2} nR(T_A - T_D) = 27713 \text{ J} \tag{24}$$

(iv) The efficiency is the total work done divided by the heat in

$$\eta = \frac{W}{Q_{AB} + Q_{DA}} = 0.28 \tag{25}$$

(v) For AB, the entropy change is

$$\begin{aligned} \Delta S_{AB} &= \int_A^B \frac{dQ}{T} = \int_A^B \frac{dW}{T} = \int_A^B \frac{P\, dV}{T} = nR \int_{V_A}^{V_B} \frac{dV}{V} \\ &= nR \ln\left(\frac{V_B}{V_A}\right) = 2R \ln 3 \end{aligned} \tag{26}$$

For BC, the entropy change is 0 because it's an adiabatic process.

For CD, the entropy change is

$$\begin{aligned} \Delta S_{CD} &= \int_C^D \frac{dQ}{T} = \int_C^D \frac{dU + dW}{T} = \frac{5}{2} nR \int_{T_C}^{T_D} \frac{dT}{T} + nR \int_{V_C}^{V_D} \frac{dV}{V} \\ &= \frac{5}{2} nR \ln\left(\frac{T_D}{T_C}\right) + nR \ln\left(\frac{V_D}{V_C}\right) \end{aligned} \tag{27}$$

Alternatively, we can use $dQ = nc_p dT$ in the first line, which will lead to the same answer.

For DA, we can find the entropy change simply by replacing the labels from the CD result

$$\Delta S_{DA} = \frac{5}{2} nR \ln\left(\frac{T_A}{T_D}\right) + nR \ln\left(\frac{V_A}{V_D}\right) \tag{28}$$

Alternatively, we can follow the same procedure as in CD, but use $dQ = nc_v dT$ instead, which will lead to the same answer.

The total entropy change should be 0 since we are returning to the same state with reversible processes, but let's calculate it as an exercise:

$$\begin{aligned}
\Delta S &= \Delta S_{AB} + \Delta S_{BC} + \Delta S_{CD} + \Delta S_{DA} \\
&= nR\ln\left(\frac{V_B}{V_A}\right) + 0 + \frac{5}{2}nR\ln\left(\frac{T_D}{T_C}\right) + nR\ln\left(\frac{V_D}{V_C}\right) \\
&\quad + \frac{5}{2}nR\ln\left(\frac{T_A}{T_D}\right) + nR\ln\left(\frac{V_A}{V_D}\right) \\
&= nR\ln\left(\frac{V_B}{V_C}\right) + \frac{5}{2}nR\ln\left(\frac{T_A}{T_C}\right) \\
&= nR\ln\left(\frac{T_C}{T_B}\right)^{1/(\gamma-1)} + \frac{5}{2}nR\ln\left(\frac{T_A}{T_C}\right) \\
&= \frac{5}{2}nR\ln\left(\frac{T_C}{T_B}\right) + \frac{5}{2}nR\ln\left(\frac{T_A}{T_C}\right) \\
&= \frac{5}{2}nR\ln\left(\frac{T_A}{T_B}\right) = 0
\end{aligned} \tag{29}$$

which is exactly what we expect it to be!

5. **Counting microstates**. Let's consider a box with $3N$ slots, each of which can hold exactly one coin. We shake the box constantly so the coins can flip sides or exchange position with its neighbors. Initially the box is partitioned into two sections, one with N slots and one with $2N$ slots. The size-N section is filled with N identical coins, and the size-$2N$ section is empty. The partition is then removed, and the N coins are free to move around the entire box. What is the entropy of the system before and after the partition is removed?

   Solution: The thermodynamic entropy is defined as $S = k\ln\Omega$. To find S, we need to count Ω, the number of possible states of the system. There are two things we need to consider:

   (1) how many distinct ways we can place the coins into available slots, and

   (2) once placed into slots, how many possible ways we can orient the ensemble of coins.

   The total number of states is the product of the two. Note that the coins are identical, but the slots are not.

   For example, if we have 2 coins and 3 slots (1,2,3), there are 3 ways we can put 2 coins into the slots: 12, 13, 23. For each way of slotting the two coins, there are 4 ways of orienting them: HH, HT, TH, TT. In total, there are 12 states.

Now we go back to our problem. Before the partition is removed, there is exactly one way of arranging the coins in the size-$2N$ section and one way of orienting the coins because there are no coins there. There is also exactly one way of slotting the N coins into all of the slots in the size-N section. However, there are 2^N ways of orienting the N coins, because each coin can take either of the two sides. As a result, there are $\Omega_{\text{before}} = (1 \cdot 1) \cdot (1 \cdot 2^N) = 2^N$ possible states.

After the partition is removed, all $3N$ slots became available for the N coins to occupy, so there are

$$\binom{3N}{N} = \frac{(3N)!}{N!(3N-N)!} = \frac{(3N)!}{N!(2N)!} \tag{30}$$

ways of slotting the coins. To see why this is the number of way to slot the coins, we can consider putting the coins one at a time: there are $3N$ options to put in the first coin, and then there are $3N-1$ options to put in the second, ... and finally there are $3N-N$ options to put in the N-th. This gives the factor $(3N)!/(2N)! = (3N)(3N-1)...(3N-N)$. However, the order with which we put in the coins doesn't matter, because the coins are identical. There are $N!$ ways of ordering the N coins, so we divide $(3N)!/(2N)!$ by $N!$ to get the above result.

For each way of slotting the coins, there are still 2^N ways of orienting them, so in total, there are

$$\Omega_{\text{after}} = \binom{3N}{N} 2^N \tag{31}$$

possible states after removing the partition.

The entropy before removing the partition is $S_{\text{before}} = k \ln \Omega_{\text{before}} = Nk \ln 2$, and the entropy after removing the partition is

$$S_{\text{after}} = k \ln \Omega_{\text{after}} = k \ln \left[\binom{3N}{N} 2^N \right] = k \ln \binom{3N}{N} + Nk \ln 2 \tag{32}$$

The change in entropy is

$$\Delta S = S_{\text{after}} - S_{\text{before}} = k \ln \binom{3N}{N} > 0 \tag{33}$$

which makes sense.